Oscillations in a charged cylinder

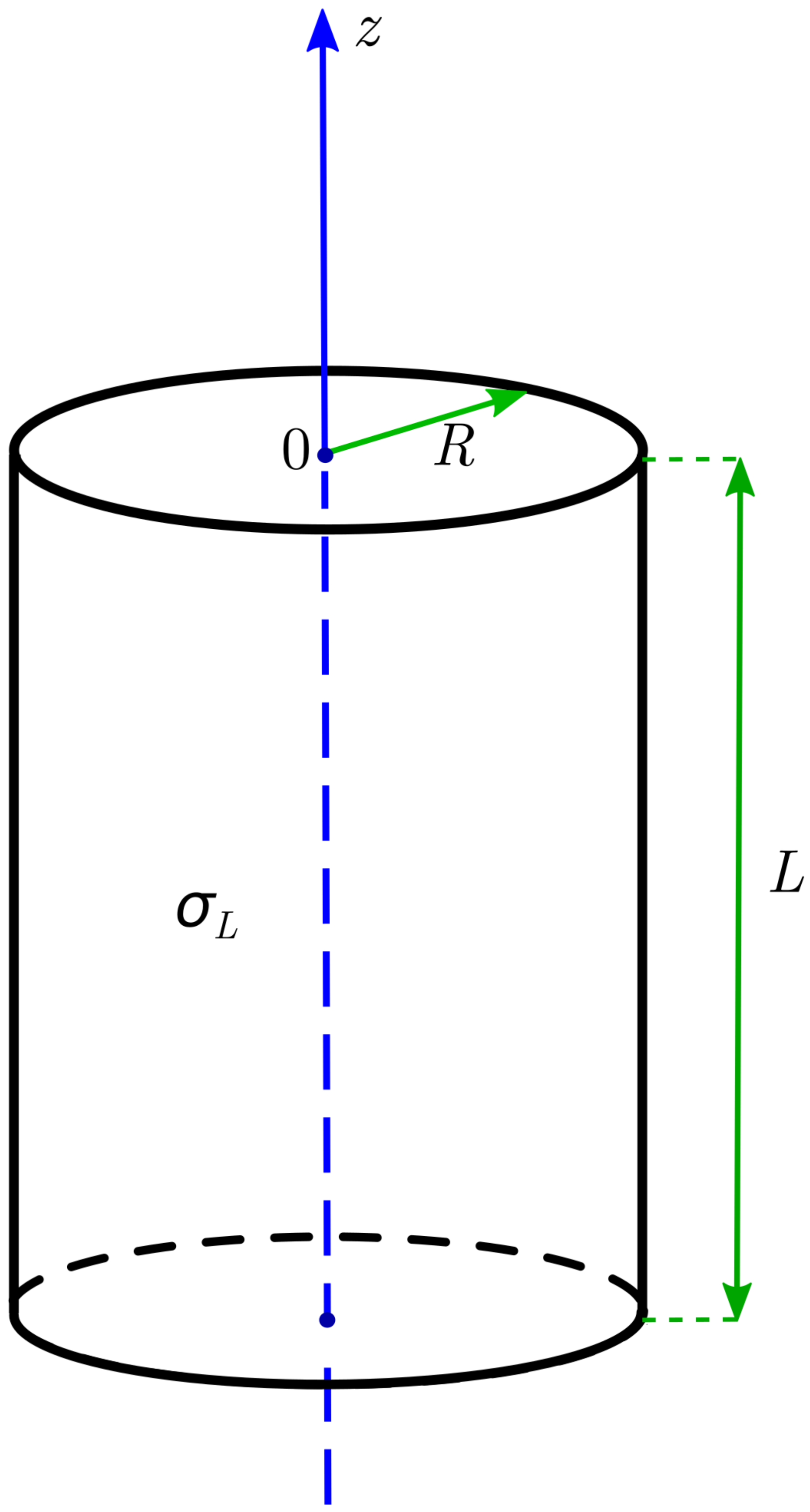
X22 (2022) — English translation

You may need the integral:

$$\int \frac{dx}{\sqrt{a^2+x^2}} = \operatorname{arth}\frac{x}{\sqrt{a^2+x^2}} + C,$$

где $\operatorname{arth} y$ обозначает обратный гиперболический тангенс числа y.

A1	A disk with radius R is charged with surface charge density σ_R. Determine the potential $\varphi(y)$ at a point on the axis at a distance y from the center of the disk. The potential is zero at infinity.	**0.50**
A2	Two such disks of radius R charged with surface charge density $\sigma_R > 0$ are parallel to each other. The distance between the centers of the disks is $2L$, the centers are located on the axis of the disks. In the equilibrium position there is a charge q with mass m, which can only move along the axis of the disks. Determine the angular frequency ω_1 of vibrations of such a charge. What is the charge sign?	**1.00**
A3	Now this charge can only move in a perpendicular direction. Express the angular frequency ω_2 of oscillations in this case in terms of ω_1. What is the charge sign now?	**1.00**
B1	The lateral surface of a cylinder with radius R and length L is charged with surface charge density σ_L. Determine the potential at a point on the axis at a distance z from the center of one of the bases of the cylinder. The potential is zero at infinity.	**1.00**



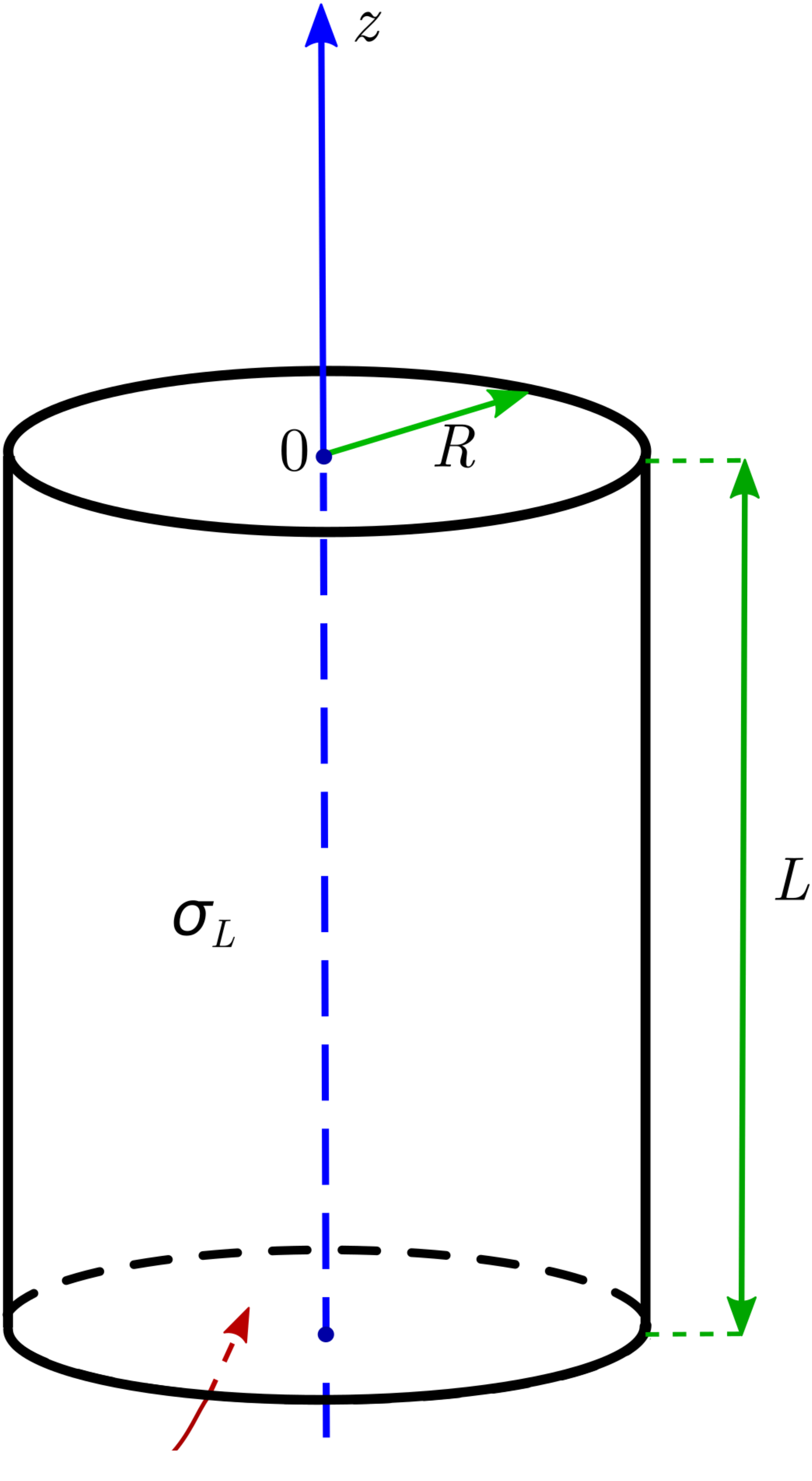
z
0
R
L
σ_L

Fig. 1.

B2	Two such cylinders (with radius R and length L, surface charged with surface charge density $\sigma_L > 0$) are placed close together and have a common axis. In the equilibrium position there is a charge q with mass m, which can only move along the axis of the cylinders. Determine the angular frequency ω_3 of vibrations of such a charge. What is the charge sign?	**1.00**

B3	Now this charge can only move in a perpendicular direction. Express the angular frequency ω_4 of oscillations in this case in terms of ω_3. What is the charge sign now?	**0.50**

C1	A charged cylinder with radius R and height $L = 40R/9$ consists of a side surface and one base. Surface charge density of the side surface σ_L, base σ_R. If you place a point charge at the center of the opposite base, it will be in an equilibrium position. Define the relation σ_L/σ_R.	**1.50**



z
0
R
L
σ_L

Fig. 2.

C2	A charged cylinder with radius $R = 28b$ and height $L = 45b$ consists of a side surface and one base. Side surface charge $\sigma_L = -8\sigma_0$, base charge $\sigma_R = 25\sigma_0 > 0$. A particle with charge $q > 0$ is placed on the axis of this system. Estimate numerically the z coordinates (in b units) of the equilibrium positions if the particle can only move along the axis. Coordinate z is calculated as in the picture. Do this as accurately as possible, however, 1% accuracy is sufficient. Answers falling within 1% of correct will receive full marks.	**2.50**
C3	Under the conditions of the previous paragraph, the particle was placed in the equilibrium position closest to the cylinder, its mass m. Determine the angular frequency ω of small vibrations of the particle.	**1.00**

Source: https://pho.rs/p/510